Ghost Predator — Auto-Scratch for Floor-Breach Fills

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The problem: the wire-race floor breach

Our entry rule says never buy above 0.75 or below 0.50. The floor is enforced three times: at signal, pre-sign, and a last-instant re-check immediately before the POST. But a FOK market BUY has no price minimum on the exchange side, and there is a ~100–400 ms gap between our last book check and the order landing. If the book reprices inside that gap, the FOK fills wherever the book now sits. We cannot close this window locally — it is wire latency. Five breaches have occurred across ~800 live trades: fills at 0.44, 0.352, 0.470, 0.300 and 0.490 — all posted while the book read ≥0.50.

Why breach fills are poison, not discounts

It is tempting to see a 0.30 fill on a side you liked at 0.55 as a bargain. It is the opposite. The only way the fill printed that low is that the book repriced against your side inside the wire gap — i.e. the market had just learned, faster than your data feed, that your side is losing. You did not get a discount; you bought someone's exit. We confirmed this live: the 0.300 breach resolved LOST. A breach fill is a position you never priced, taken at the exact moment informed flow was leaving it. Treat it as adverse selection by definition.

Breach history (B's book)

Fill price	Floor at the time	Outcome	Note
0.440	0.50	(pre-fix era)	first observed breach
0.352	0.50	(pre-fix era)	led to the 3-layer floor checks
0.470	0.50	WON	marginal breach, got lucky
0.300	0.50	LOST	the smoking gun — informed reprice
0.490	0.50	WON	hair under floor; happened during patch work

Takeaway: 2W–1L on resolved breaches looks survivable, but the wins were luck on coin flips we never chose to bet. Expectation is negative once the informed-flow mechanism is understood.

The fix: auto-scratch (sell it back immediately)

When a fill comes back below FILL_FLOOR, the bot now immediately posts a FOK **SELL** of the entire position at a limit of (fill price – 0.15). Two possible outcomes, both clean:

• **Sell fills** → fully out within ~1 second of entry. Cost is roughly the spread (typically $0.30–$1.50 on a $10 stake) instead of holding a $10 coin flip. The row is recorded with status `scratched` and the round-trip PnL.
• **Sell kills or errors** → the bot holds the position exactly as before the fix. Fails safe to the status quo; no partial states, no new failure modes.

Implementation notes (replicate carefully)

1. **FOK, not FAK, on the scratch sell.** A FAK can partially fill and leave you holding a fraction of a position your resolver does not understand. FOK is binary: fully out or fully held.
2. **The scratched status must be invisible to your ladder.** Our paroli stake query only reads won/lost/open rows, so a scratch neither steps nor resets the ladder — correct, because it is a trade you never priced. Check your own ladder query before copying.
3. **Resolver compatibility.** Anything that picks up status='open' rows (resolver, provisional verdicts, exit loggers) automatically ignores scratched rows. Verify yours does too.
4. **The detection already exists** if you log floor breaches: trigger the scratch in the same branch that prints your breach alert, after shares/fill_price are known.
5. **Keep all pre-POST floor checks.** The scratch is the fourth layer, not a replacement — it handles only what the first three physically cannot (the wire gap).

Config

```
AUTO_SCRATCH=true    # enable the sell-back
SCRATCH_MAX_SLIP=0.15  # worst acceptable slip vs the breach fill price
```

Expected frequency: rare (~5 in 800 trades, clustered in fast tape). Expected effect: converts each breach from an unpriced ~$10 coin flip into a ~$1 scratch. The fix changes nothing about normal fills.